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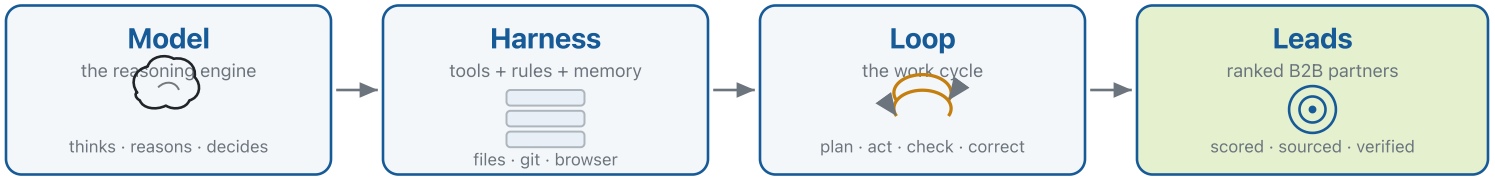
# Agentic AI for Lead Generation

## Partner companies for the chair

Tobias Weiß · Guest Lecture · Professur Marketing & Verkaufsmanagement

Thursday, 22 October 2026 · 15:15–16:00
within "Advanced Issues in Marketing and Sales" (Prof. Haas)

45 minutes · BWL Master students · WS 2026/27 edition



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# Why this session exists

- The chair partners with companies: funding, data, cases
- Finding + qualifying partners is manual, slow, unscalable
- Agentic AI can research, score and rank prospects
- **Demo output = evidence for "Sales Strategy and Execution" (SS 2027)**

> Nobody wins from another lecture about chatbots.

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# What you will be able to do

| By the end… | You can… |
|---|---|
| **Understand** | Explain what "agentic" means, and why it is not a chatbot |
| **Operate** | Read how a harness (pi) gives an AI tools, rules and a task |
| **Assess** | Judge an agentic output: checked? citable? repeatable? |
| **Apply** | Run the exercise yourself: ingest + rate leads for the chair (repo → rubric → validator) |
| **Context** | Say which context an agent can't know — and why that keeps you in charge |

> Model = brain. Harness = hands. Workflow = muscles. **Context = memory — the part only you can supply.**

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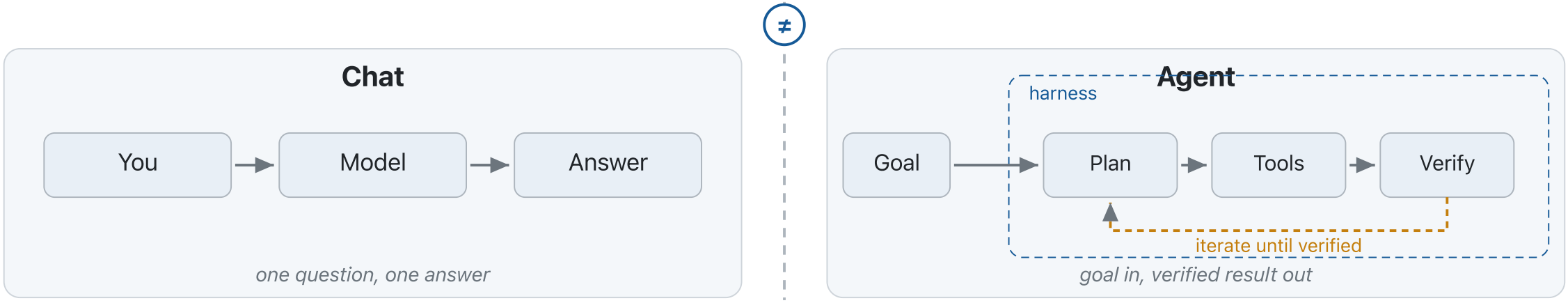
# Roadmap — 35 min + 10 min Q&A

| Block | Content | Time |
|---|---|---|
| 1 · Foundations | agentic ≠ chat · model · harness · loop · providers · corpus & rate limits · git & the repo | ~11 min |
| 2 · Method | contracts · the lead-gen loop · rating formulas · evidence | ~8 min |
| 3 · Walkthrough | live demo: repo → agent → validator | ~13 min |
| 4 · Reflection | take-aways · resources | ~4 min |
| 5 · Q&A | your questions · discussion | 10 min |

> The walkthrough is the heart — theory is fuel for it. **10 minutes are reserved for your questions.**

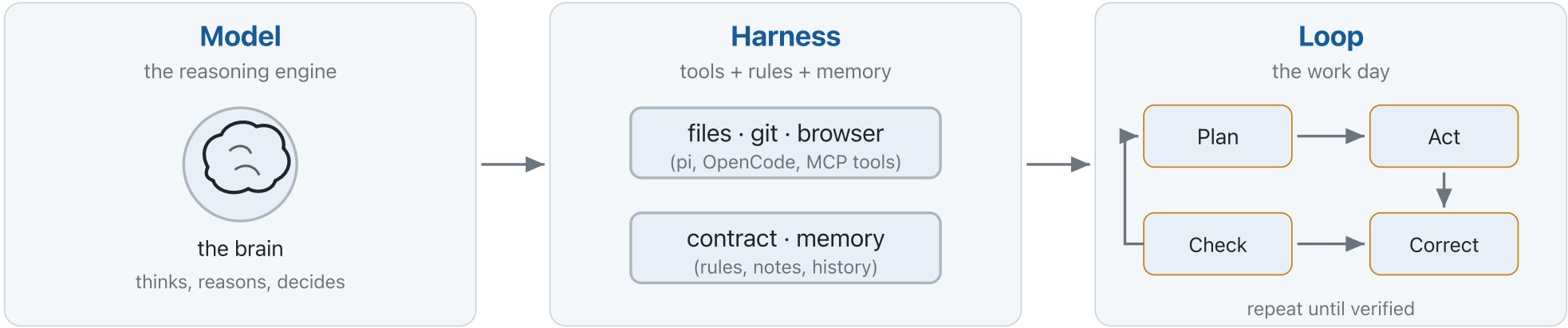
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# Agentic ≠ chat




> "Agentic" is a working environment — not a better chat.

# The parts: model · harness · loop




**Harnesses (2026):** pi, OpenCode, Claude Code — same model, different results. The harness decides.

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# Key AI providers — what you can use as a student

| Provider / offer | What you get | Where |
|---|---|---|
| **JLU HRZ API-Service** ★ | Free LLMs for JLU members: local Qwen (256K–1M ctx) + commercial | uni-giessen.de → HRZ → KI · **ki@uni-giessen.de** |
| **OpenCode Zen — free models** | **Big Pickle** (stealth) · **Beam Free** (Reflection 501B, 1M ctx, no training) | opencode.ai/zen — `/models` |
| **Google AI student plan** | 1 year Google AI **Plus** free · Gemini, 400 GB | one.google.com/ai-student |
| **OpenRouter** | one key → many models (OpenAI, Anthropic, Google, open) | openrouter.ai |
| **Harnesses** | today's demo tools — **OpenCode**, pi | opencode.ai · pi.dev |
| **Local models (Ollama)** | free, on your laptop, no account, private | ollama.com |

> ⚠️ **Privacy:** Big Pickle's provider **may train on your input**; Beam Free and HRZ local models **do not**. **Never paste private, customer, or unpublished data into a model that trains on user input** — for sensitive work use HRZ or Ollama. · Start with **HRZ API**.

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# The corpus and the rate limits — know your scale, know your ceiling

The agent searches a corpus of **>14,000 marketing papers** (title, authors, abstract, venue, URL for each) and calls the model in a loop — a lead-gen pass is dozens of calls. So the rate limit matters.

| Provider | Limit | Notes |
|---|---|---|
| **HRZ API-Service** ★ | free for JLU · no published cap | local Qwen, 256K–1M ctx |
| **OpenCode Zen** (free models) | ~15–20 req/min | silent 429 · models are temporary |
| **Google AI student plan** | ~90 prompts/day (Gemini 3 Flash) | AI Plus 2× standard · refreshes every 5h |
| **OpenRouter** | free tier limited | pay-as-you-go lifts the cap |
| **Ollama** (local) | no API limit | bounded by your hardware (VRAM, tokens/sec) |

> Free tiers are fine for learning. For a real run, start with **HRZ** — free, local, no training on your data.

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# The workspace: git and the repo itself

## **git** = a lab notebook that never forgets

| This repo — `marketing-research`, public | The numbers |
|---|---|
| The **corpus**: >14,000 papers in one `papers.yaml` — title · authors · abstract · venue · URL | **>150 commits** since Aug 2026 · 1 researcher + AI agents · >90 last month |
| The **machinery**: taxonomy · discovery scripts · **executable validators** | generated files are **never hand-edited** |
| **Your exercise**: `leads/` — same rules, same history | every lead a commit · every claim a URL |

`github.com/tobias-weiss-ai-xr/marketing-research`

> If it isn't in the repo, it didn't happen. If it is, you can check it — line by line, back to day one.

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# Why now — the evidence inside the research corpus

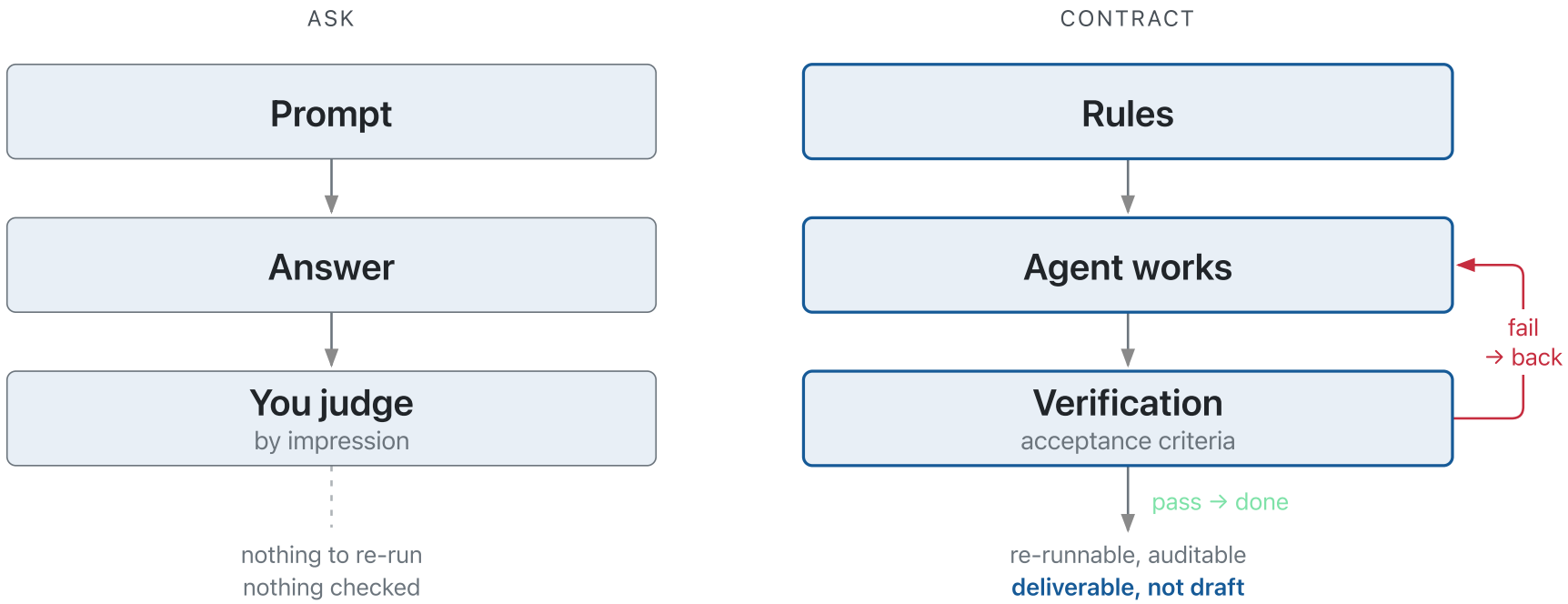
The chair's open corpus, public on GitHub: **>14,000 marketing papers — >8,700 published in 2026 alone**

| Signal | Count |
|---|---|
| Papers mentioning **agentic AI** | >100 — >90 of them from 2026 |
| Papers on **lead generation** | >30 |
| **Both** — strict (agentic × lead-gen) | **≥ 2** |
| **Both** — broader (agent-aware × lead-ops) | **≥ 5** |
| Papers on **context-aware** marketing | >20 |

> Two rounds of OpenAlex ingestion took the strict intersection from **1** to **≥ 2**, the broader set from **3** to **≥ 5**. The field just opened — claim the gap while it's a gap.

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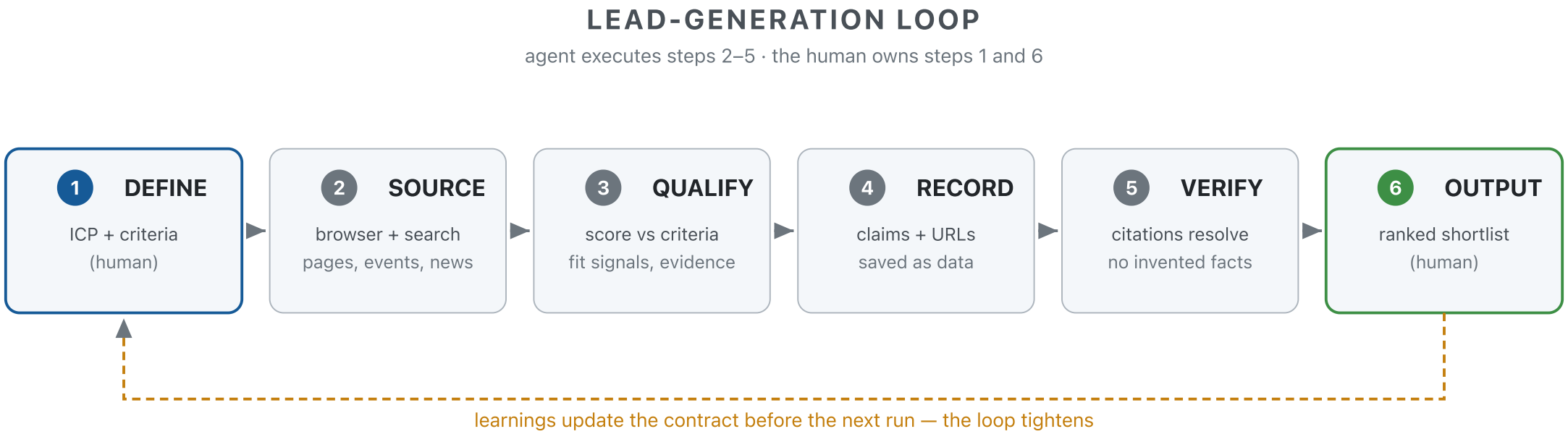
# How to talk to the machine: prompts vs contracts




Prompt = asks a chat · Contract = runs an agent · **Context = what the machine can't know**

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# A lead-generation loop as an agentic workflow




Six steps — the agent executes 2–5, you own definition and evaluation. **Those two boxes are where context lives.**

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# Define the hunt: scoring the sector

Lead generation starts with a **human** decision: *which sector do we hunt?*

| Criterion (1–5) | Mech. & plant engineering | Optics / photonics | Grocery retail |
|---|---|---|---|
| Chair research fit — sales management | 5 | 5 | 2 |
| Course fit — "Sales Strategy and Execution" | 5 | 5 | 2 |
| Regional employers (Mittelhessen) | 5 | 5 | 3 |
| Lead-gen relevance — B2B, long cycles | 5 | 4 | 2 |
| Demo data availability | 4 | 4 | 5 |
| **Total** | **24** | **23** | **14** |

> Hard guard: sector is out if chair fit < 4 **or** lead-gen relevance < 4 — grocery retail (our first default) fails. **Decision: German mechanical & plant engineering.**

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# Rate a lead: pick a formula — before the agent runs

| Formula | Rule | Trade-off |
|---|---|---|
| Weighted sum | `L = Σ wᵢ·sᵢ` · w: fit .35 · region .25 · size .15 · evidence .15 · timing .10 | transparent · **compensatory**: a 5 on fit can hide a 0 on evidence |
| Geometric mean | `L = Π sᵢ^wᵢ` | **non-compensatory**: one zero sinks the lead · harsh to defend |
| **Gated weighted sum** ✓ | gates first — B2B · region reachable · ≥ 1 verified source — then weighted sum | matches the chair's decision · thresholds = judgement calls |

Example (each criterion 0–5): fit 5 · region 3 · size 4 · evidence 4 · timing 2 → `L = 3.90`

> Weights live in the repo ( `leads/leads.yaml` , `rubric:` ) — **published context, not vibes**. Change them — in a PR, with reasons.

# Now: a live walkthrough

Task: ingest **2 leads** into the chair's repo — `leads/leads.yaml` — with:

- scores per criterion + the weighted sum, gates applied, ranked
- named source URL per claim (citable, verifiable)
- `python3 leads/validate_leads.py` passing before anything counts

I run this live in a harness — in the same repo you will work in.

No fallback video: this exact exercise was **tested agentically**, end to end.

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# Walkthrough · Step 1 — The repo is the contract

```
marketing-research/                  github.com/tobias-weiss-ai-xr/marketing-research
├─ papers.yaml                       14,088 papers — single source of truth
├─ config/taxonomy.yaml              categories + discovery queries
├─ scripts/validate_papers.py        the contract, executable
├─ leads/leads.yaml                 your starting point: schema + rubric
├─ leads/validate_leads.py          same pattern: validate before commit
└─ README.md · statistics.json       auto-generated — never hand-edited
```

> No prose rules file the agent can ignore: **structure + validators + one source of truth.**

# Walkthrough · Step 2 — The agent at work

Watch the six-step loop run live:

- **only watch** — no magic, every tick is visible in the terminal
- agent reads the **repo** (schema + rubric), then **opens the browser** (MCP) — real company pages
- **why a browser:** LinkedIn, consent walls, logged-in portals block plain HTTP — the good context sits behind them
- writes leads into `leads/leads.yaml` → runs the **validator** → fixes what fails
- **iterates**: pause, correct, re-check — that is the loop

> Nothing happens off-screen — the loop is fully visible.

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# Walkthrough · Step 3 — The three questions

1. **Checked?** Did the agent verify its output, not just generate it?
2. **Citable?** Does every claim have a URL that resolves — and actually supports it?
3. **Repeatable?** Same repo, same rubric, same validator → a result you can re-run and audit?

*The 1–5 scores are a transparent heuristic — each criterion with evidence — not a measurement.*

> All three guard one thing: **context** — invented context (Checked) · untraceable context (Citable) · lost context (Repeatable).

> The final slide of the walkthrough: apply them to the live result.

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# Your turn: clone the repo, ingest 2 leads

```
# Fork on GitHub, then clone your fork:
git clone https://github.com/YOUR-USERNAME/marketing-research.git
cd marketing-research
# … edit leads/leads.yaml, then validate …
python3 leads/validate_leads.py
git add leads/leads.yaml
git commit -m "Add 2 leads: CompanyA, CompanyB"
git push   # → open a pull request on GitHub
```

1. **Fork & clone** — the commands above (repo is public)
2. **Ingest 2 leads** — real B2B companies, every claim with a URL
3. **Score & rank** — gates first, then the published formula
4. **Validate & push** — validator passes → `git push` → pull request

> Best submissions join the chair's real shortlist — "Sales Strategy and Execution", SS 2027.

# What the chair gets out of it

- A reproducible partner-sourcing method (not a one-off search)
- A ranked, sourced shortlist — fed by the course's best submissions
- Students who can run agentic workflows (a real skill for the CV)
- Evidence base for the **Bachelor course "Sales Strategy and Execution"** (SS 2027)

> The lecture is not the output. The shortlist is.

# Take-aways

1. Agentic = working environment, not a chatbot
2. Harness = model + tools + rules = accountable AI work
3. The repo is the contract: schema, rubric, validator — "done" is executable
4. Verify, cite, repeat — the three questions
5. **Context beats compute — the right context contract beats a bigger model**

> Next summer semester, the chair decides what this feeds: the Bachelor course "Sales Strategy and Execution".

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# Resources

- **JLU HRZ API-Service** (free LLMs for JLU members) — uni-giessen.de → HRZ → KI → API-Service · model overview
- **HRZ access one-pager** (student quick start, 1 page) — `hrz-access-one-pager.md`
- **pi** — pi.dev · docs: pi.dev/docs · **OpenCode** — opencode.ai · **OpenRouter** (one key, many models) — openrouter.ai
- **Google AI student plan** — one.google.com/ai-student (1 year free for students)
- **marketing-research** (chair corpus, public) — github.com/tobias-weiss-ai-xr/marketing-research → start at **`leads/`** (exercise brief · rubric · validator)
- **skeleton-research** (forkable corpus skeleton) — github.com/tobias-weiss-ai-xr/skeleton-research
- **This deck** — `lecture/` in this corpus repo (English, 45 min)

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# Thank you!

## Questions?

Tobias Weiß · guest lecture · Professur Marketing & Verkaufsmanagement

Agentic AI for Lead Generation · 22 Oct 2026 · 45 min